

Attendance ——— 20

Lecture-1
19.07.2026

CT/presentation ——— 20
20
Assignment ——— 20
40

Mid ——— 24

Final ——— 36
100

g222 CT - Lecture - 2, 2, 3

Basic Discussion

Lecture-2
20.07.2026

Component of microcomputer system;

There are three parts of computer system.

1. Central Processing Unit (CPU): The CPU controls the operations of the computer and executes all instructions. It is often referred to as the brain of the computer. Two major sub-units carry out through CPU - processing and management. Summarized For CPU:

⇒ Function: Controls the ~~over~~ overall operation of the computer

⇒ Execution: Processes and executes all instructions received by the system.

⇒ Role: Serves as the main processing core and is commonly referred to as the brain of the comp.

2. Memory: Memory serves as the main storage unit of the system, retaining the data and instructions required for processing.

⇒ Function: Serves as the primary storage area within the computer system.

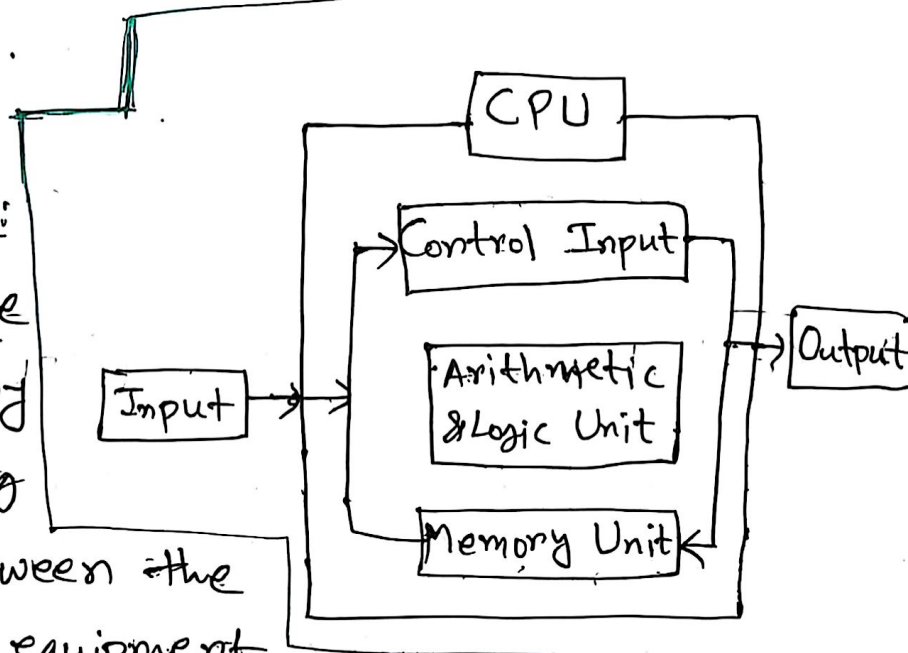
⇒ Role: Holds and stores both data and instructions needed for processing.

3. Input/Output Devices:

Input/Output devices handle system communication by ~~transforming~~ transferring data back and forth between the computer and external equipment or user.

⇒ Function: Enables communication between the computer and external environment.

⇒ Role: Transfer data back and forth between the computer system and external devices.



Q. Why CPU is called the brain of the computer?

⇒ The central processing unit (CPU) is commonly called the brain of the computer for two main reasons based on how it functions within a microcomputer systems:

1. Controls System Operation: It directs and coordinates all operation across the computer, managing how components work together.

2. Executes Instructions: It processes and executes all instructions received by the computer system.

Inside CPU, the Control Unit (CU) manages the ~~fetch~~ ~~fetch~~ fetch-decode-execute cycle, while the ALU carries out actual calculations and data transformation.

Operation of Control Unit

1. Fetch: Control unit ~~fetch~~ fetches instructions from memory that need to be executed.

2. Decode: After fetching, control unit decodes the instruction. Decode means understand the binary pattern.

3. Execution: After decoding, control unit executes the instructions.

Operation of Arithmetic Logic Unit (ALU)

1. Arithmetic Operation: Arithmetic operation like Addition, Subtraction, Multiplication etc. is performed here.

2. Logical Operation: Logical operation like AND, OR, NOT is performed here.

3. Data Manipulation: Altering or transforming data to make it more meaningful and manageable.

Microprocessor

Definition. Microprocessor is a processor which incorporates the functions of a CPU on a single Integrated Circuit (IC).

Types (1) 8086 (2) 8085 (3) 8088

Application: Microprocessors are integrated circuits designed for big, multi-purpose, general-purpose applications where high computing power, flexibility and dynamic multitasking are needed. Examples:

1. Personal Computer: Used as the core processing engine (CPU) for desktop and laptop computers running complex operating systems, softwares and heavy computational tasks.

2. Mobile Device: ^{Used for in} Power smartphones and tablets to handle multi-tasking applications, user interfaces, graphics processing and web browsing.

Microcontroller

Definition: Microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system.

↓
Embedded System: Embedded systems are focused on controlling and monitoring specific functions within a device.

Application Microcontrollers are compact ICs designed for embedded systems. Their primary purpose is to execute a single, dedicated task efficiently by utilizing their built-in processor, internal memory, and internal I/O components.

① Washing Machine: Controls and monitors specific dedicated functions such as water levels, wash/spin cycles, timing and temperature.

② Fire detection and safety Devices: Monitors sensors continuously to detect smoke or rapid temperature rises and triggers alarm or safety measures instantly when thresholds are met.

Difference

Microprocessor	Microcontroller
1. It is used for big applications.	1. It is used to execute a single task
2. It is the heart of computer system	2. It is the heart of the embedded system.
3. It is just a processor; Memory and I/O components have to be connected externally	3. It contains external purposes processor along with internal memory and I/O components
4. Power consumption is high	4. Power consumption is low
5. Cost is more	5. Cost is less
6. Difficult to replace	6. Easy to replace
7. Application: Personal Computer	7. Application: Washing Machine

Microprocessor

Lecture-3

~~23.07~~

26.07.2020

Generation of Microprocessors:

First Generation (betⁿ 1971-1973)

- ⇒ PMOS technology
- ⇒ 4 bit processor (16 pins)
- ⇒ 8 and 16 bit processors (40 pins)

Second Generation (During 1973)

- ⇒ NMOS technology
- ⇒ Faster Speed, Higher density
- ⇒ 4/8/16 bit processors (40 pins)
- ⇒ Intel 8085 (8 bit processor)

Third Generation (During 1978)

- ⇒ HCMOS technology
- ⇒ Faster Speed, Higher Packing density.
- ⇒ 16 bit Processors (40/48/64 pins)
- ⇒ Easier to program
- ⇒ Dynamically relatable programs.
- ⇒ Intel 8086 (16 bit processor)

Fourth Generation (During 1980)

- ⇒ HCMOS technology
- ⇒ 32 bit processors.
- ⇒ Physical memory space
2²⁴ bytes = 16 Mb
- ⇒ Virtual memory space
2⁴⁰ bytes = 1 Tb
- ⇒ Intel 80386

Fifth Generation

Pentium

Pentium (1993)

- ⇒ 32 bit microprocessor
- ⇒ 64 bit data bus
- ⇒ 32 bit address bit
- ⇒ 4 GB main memory
- ⇒ 60,66,90 MHz

Pentium Pro (1993)

- ⇒ 32 bit microprocessor
- ⇒ 64 bit data bus.
- ⇒ 36 bit address bus.
- ⇒ 64 GB main memory
- ⇒ Starts at 150 MHz.

Pentium II (1997)

- ⇒ 32 bit microprocessor
- ⇒ 64 bit data bus
- ⇒ 36 bit address bus
- ⇒ 64 GB main memory
- ⇒ Starts at ~~226~~ 266 MHz

Pentium III (1999)

- ⇒ 32 bit microprocessor
- ⇒ 64 bit data bus
- ⇒ 36 bit address bus.
- ⇒ 64 GB main memory
- ⇒ Starts at 450 MHz.

Pentium iv (2002)

- ⇒ 32 bit microprocessor
- ⇒ 64 bit data bus
- ⇒ 92 bit address bus
- ⇒ 4 GB main memory
- ⇒ starts at 1.4 GHz.

→ A summary table of Generation of Mic.

↳ check Lecture-3 (page -8)

8080/8085 Microprocessor

⇒ 8 bit microprocessor

⇒ Developed by INTEL in the mid 1970

⇒ NMOS technology

⇒ Address bus 16 bit. $\left\{ \begin{array}{l} 2^{16} = 65536 \text{ byte or } 64 \text{ KB} \\ [65536/1024 = 64] \end{array} \right\}$

⇒ Data bus 8 bit. $\rightarrow$ (Data bus 20 bit processor 8 bit)

⇒ Non-pipelined processor.

⇒ It has 3 available clock speeds (3 MHz, 5 MHz, 6 MHz)

⇒ 8085 has memory capacity of 64 KB

↳ $2^{16} \text{ Bytes} = 2^6 \times 2^{10} \text{ Bytes}$
 $= 64 \times 1 \text{ KB}$
 $= 64 \text{ KB}$

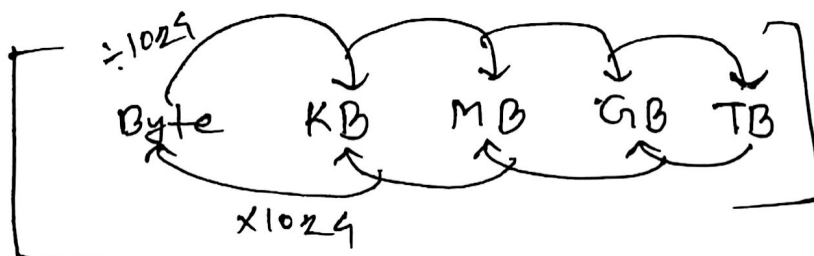
$\text{KB} = 2^{10}$
$\text{MB} = 2^{20}$
$\text{GB} = 2^{30}$
$\text{TB} = 2^{40}$

⇒ It does not have an instruction queue.

8086 Microprocessor

- ⇒ 16 bit ~~micro~~ microprocessor
- ⇒ Developed by INTEL in the year 1978. technique.
- ⇒ Originally HMOS, now manufactured using HMOS IIIA
- ⇒ Approximately 29,000 transistors and 40 pin.
- ⇒ 5V supply.
- ⇒ 33% duty cycle.
- ⇒ Address bus 20 bit. $\left\{ \begin{array}{l} 2^{20} = 2 \times 2^{20} \text{ Bytes} \\ = 2 \times 1 \text{ MB} \\ = 2 \text{ MB} \end{array} \right\}$
- ⇒ Data bus 16 bit
- ⇒ Pipelined processor
- ⇒ It has 3 available clock speeds (5 MHz, 8 MHz and 10 MHz).
- ⇒ 8086 has memory capacity of 1 MB
- ⇒ It has 6 byte instruction queue.

*
Q) What is 16-bit Processor: A processor that can process 16-bit data at a time is called a 16-bit processor.



$$\frac{A}{1024^n}$$

Difference between 8085 and 8086 Microprocessor

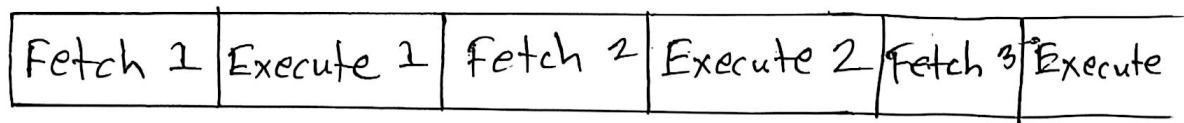
8085	8086
8 bit microprocessor	16 bit microprocessor
Address bus 16 bit	Address bus 20 bit.
Data bus 8 bit	Data bus 16 bit
Non-pipelined processor	Pipelined processor
Memory capacity 64 KB	Memory Capacity 1 MB.
NMOS Technology	HMOS III technology

Pipelining in microprocessors

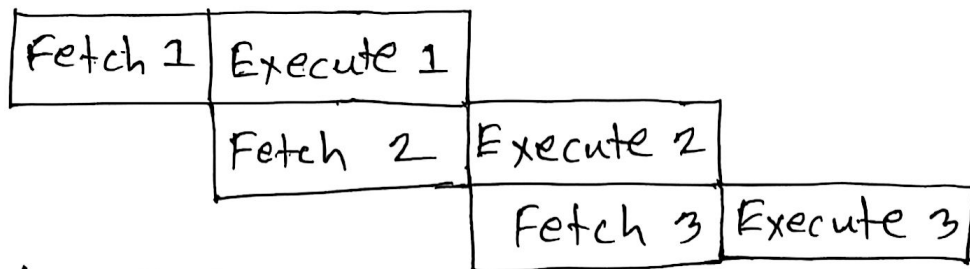
The process of fetching the next instruction when the present instruction is being executed is called as pipelining.

Types:

Non-pipelined:



Pipelined:



Advantages:

- Requires less time.
- Speed fast
- Efficiency is more

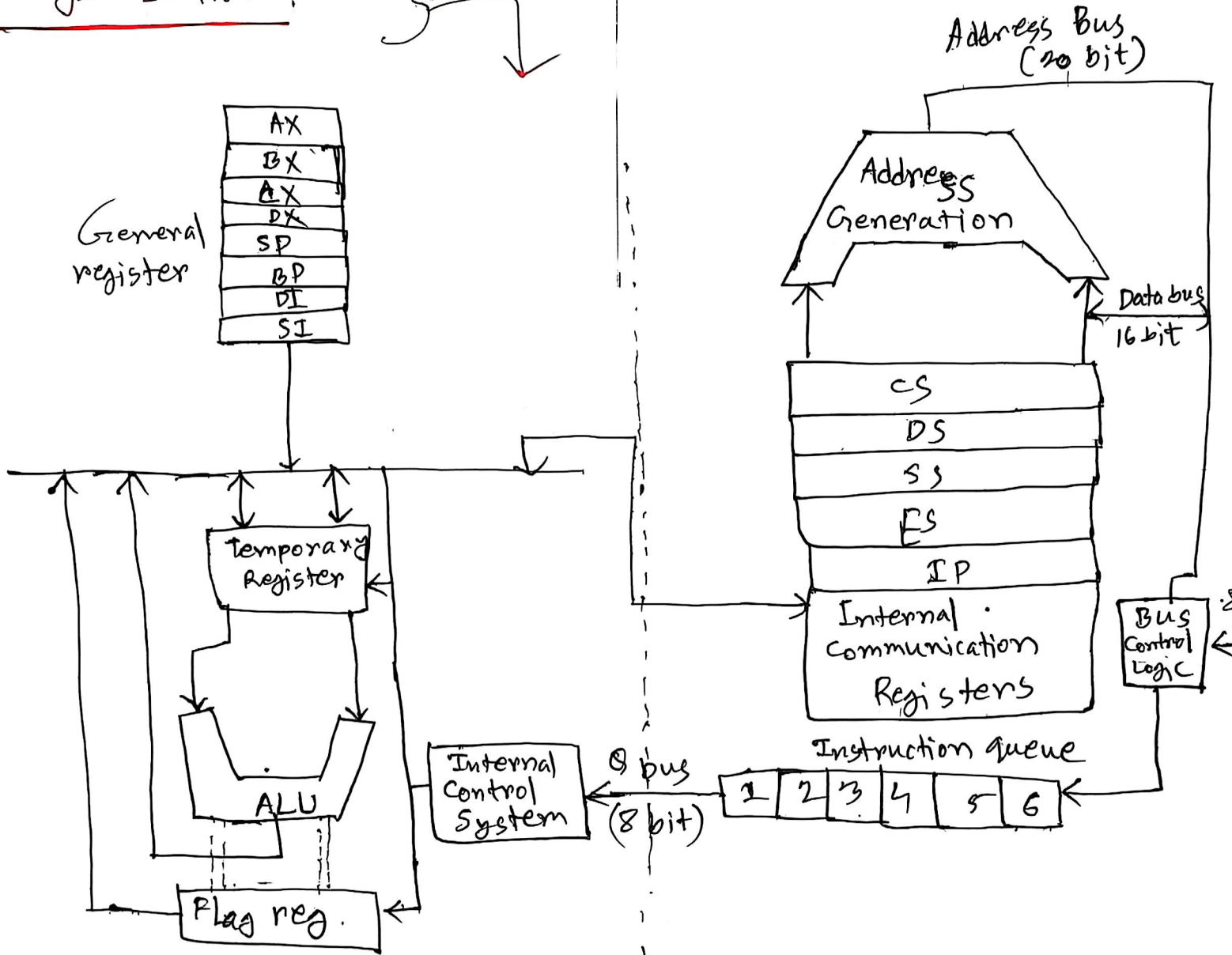
8086 Microprocessor

Lecture-4

27.07.2026

Internal Block Diagram:

Organization:



Execution Unit (EU)

- Instruction decode and execute
- Arithmetic and logical operation

Bus Interface Unit (BIU)

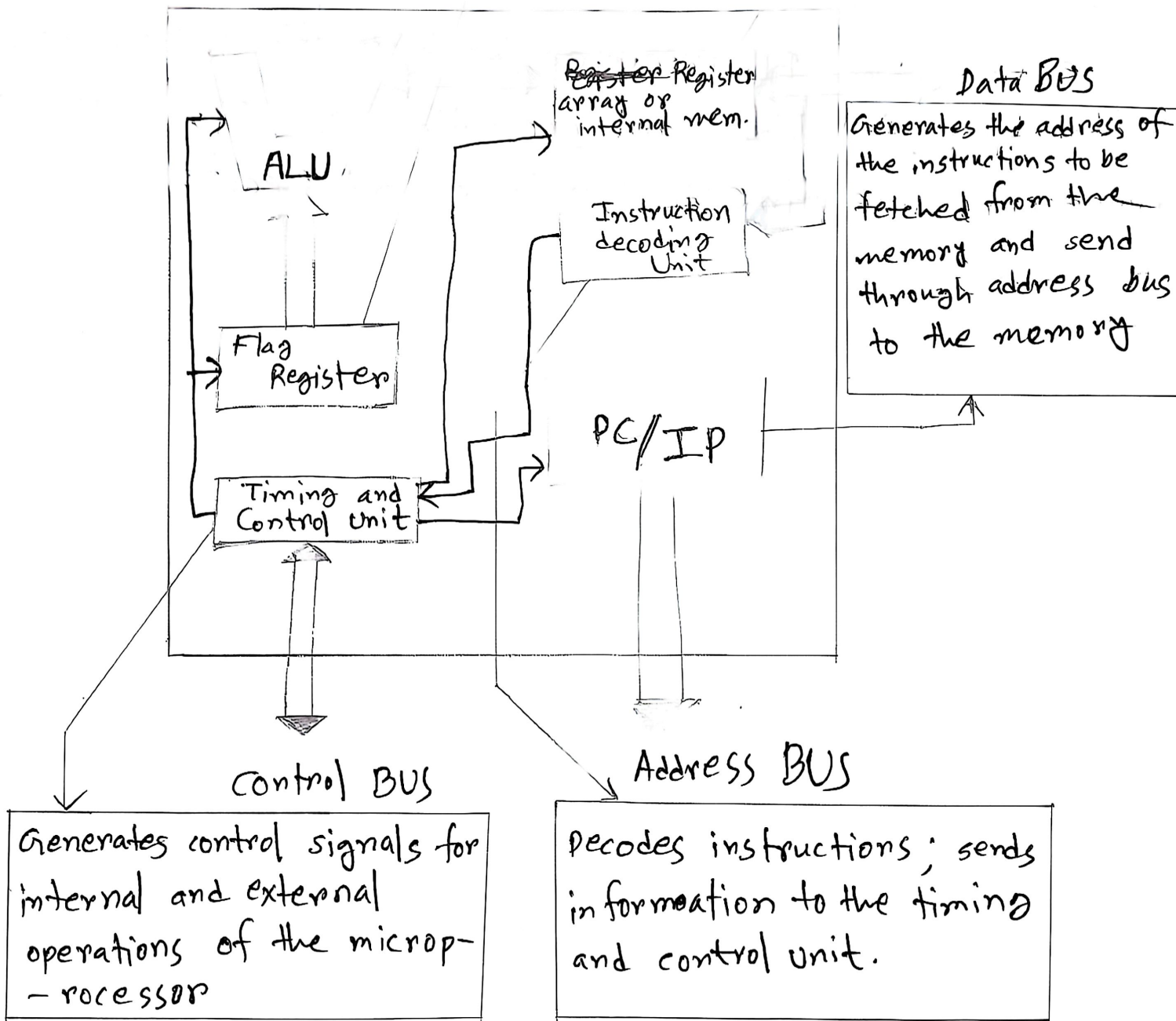
- Instruction fetch.
- Address generation
- Memory and I/O communication.

Functional Block Diagram.

Computational Unit:
Performing ~~the~~ arithmetic
and logic operations.

Various condition of the
result are stored as
status bits called
flags in flag register

Internal storage of data



Q. How does queue speed up the processing of the 8086 microprocessor?

⇒ The 8086 microprocessor's 6-byte instruction queue increases speed through instruction prefetching. The BIU fetches instructions and stores them in the queue. At the same time, the EU executes the instructions. Because of instruction fetching and execution overlap, the processor's idle time is reduced and execution speed increases.

Register of 8086 Microprocessor

Lecture-5

02.08.2026

AX → AL AH

Data ରେଖି AX

BX → BL BH

" ରେଖି " [AL, AH]

CX → CL CH

କ୍ଷେତ୍ର AL ଏବଂ AH
↓
low High

DX → DL DH

ସମୀକ୍ଷା: ଏହି topic Lecture-4 (slide) page (7-20) ମଧ୍ୟରେ ଅଛି।

Assembly Language

Lecture-6

03.08.2026

Syntax

MOV
↓
opcode
↓
operation

CX, 5
↓
operand

~~MOV~~ initialize counter
↓
comment

AX, BX
↓ source
↓
destination

⇒ Hexa-decimal -এ ৭টি অঙ্ক আছে। n/A দিলে হবে এর sequence
-এ শুরু দিলে numerical না থাকলে আশ্রিত 0 দিলে হবে। যেমন:

1FFH | 0 লাগবে না

0ABFF → 0 লাগবে আশ্রিত

যাকি Lecture-5(slide) page(1-17) পড়তে হবে।

Lecture-5(slides-5)

page(18-38)

Lecture-7

17. ~~10~~ 08.2026

Class lecture

MOV, AX, BX →

Before	
AX	BX
01	02

After

AX	BX
02	02

XCHG, AX, BX →

Before	
AX	BX
01	02

After

AX	BX
02	02

$a = b$

shortcut of 2's complement

~~MOV a, b~~

(MOV AX, b)

MOV a, AX

000100100
111011100

110010001
↓
001101111

0002 → FFFE

0000 0000 0000 0010
1111 1111 1111 1110
F F F E

ସର୍ବପ୍ରଥମ ଡାଇଟିକ
124କ ଖୁବ୍ ବଡ଼
ହେବ ।

ଅନ୍ତର ୦/୫ first '1'
ମାତ୍ରା ମଧ୍ୟ ସାମେ
ଲିଖା ହେବ ।

ଅନ୍ତର ବାକି ଗୁଣା ହେବ
ହେବ ହେବ

Page-38

$B = A$

↓

MOV AX, A ;

MOV B, AX ;

$A = B - 2 \times A$

↓

MOV AX, B

SUB AX, A

SUB AX, A

MOV A, AX

$A = 5 - A$

↓

MOV AX, 5

SUB AX, A ;

MOV A, AX

XB	XA
50	10

← XB, XA, VOM

7	XA
0	0

XB	XA
50	10

← XB, XA, VOM

Lecture-5 (slide-5)

page (39-44).

Lecture-8

23.08.2026

$a \rightarrow 17$

$A \rightarrow 65$

Output: 32

Chapter-5 (slide-6)

The Processor Status and the Flag Register

(Slide + Lecture পাঠ্য শব্দ)

Carry Flag

- i) 8/16 bit যার জন্য যেকোনো ক্যারি বাহ্যে আসবে হবে।
- ii) অর্থাৎ bit-এ operation করার পর যদি carry থাকে তাহলে $CF = 0$ অন্যথায় $CF = 1$.

Parity Flag

- i) Parity 16 bit হলে প্রথম 8-bit consider করে হিসাব করতে হবে।
- ii) 8-bit হলে 8-bit হিসাবই হিসাব করতে হবে।
- iii) Operation করার পর result অব 0 হলে $PF = 1$
- iv) result অব 0 না হলে কতগুলো '1' আছে count করতে হবে।

যদি count এর ফলাফল even হয় তাহলে $PF=1$ অন্যথায় $PF=0$ হবে।

Auxiliary Flag

- i) AF নিয়ে কাজ করার middle part টুকু নিয়ে কাজ করতে হবে।
- ii) 16-bit হলে 8-bit বই কাজ করতে হবে আর 8-bit হলে 4-bit বই কাজ করতে হবে।
- iii) Middle-point bit-এ operation এর পর যদি carry থাকে তবে $AF=1$ অন্যথায় $AF=0$.

$$\begin{array}{r} 10101101 \\ 00110010 \\ \hline 00000000 \end{array} \rightarrow \text{carry নাই তাই } AF=0$$

Zero flag

Operation করার পর প্রাপ্ত result 0 হলে $ZF=1$; নতলে

$$ZF=0$$

Sign flag

সর্বোচ্চ bit যদি 1 হয় তাহলে $SF=1$ নতলে $SF=0$

Signed overflow

- i) ଯଦି ସଂଖ୍ୟା ସମାନ ସିଗ୍ନ ସହିତ ଫଳାଫଳ ସଂଖ୍ୟା ବିଭିନ୍ନ ସିଗ୍ନ ଥାଏ ତେବେ signed overflow.
- ii) ଯଦି ଦୁଇଟି ବିପରୀତ ସିଗ୍ନ ସଂଖ୍ୟା ଯୋଡ଼ା ଫଳାଫଳ ସଂଖ୍ୟା ସମାନ ସିଗ୍ନ ଥାଏ ତେବେ signed overflow.

Signed Overflow Maths

Lecture-9

30.08.2026

For OF (overflow flag) value is either 1 or 0 on the basis:

- i) Positive + Positive = Negative
 - ii) Negative + Negative = Positive
 - iii) MSB in $\neq$ MSB out
- } OF = 1

Math-1: 41h, 4Fh {Add}

$$\begin{array}{r} \text{MSB out} \quad \text{MSB in} \\ 41h = 0100\ 0001 \\ 4Fh = 0100\ 1111 \\ \hline 1001\ 0000 \end{array}$$

$$AF = 1$$

$$CF = 0$$

$$PF = 1$$

$$SF = 1$$

$$ZF = 0$$

$$OF = 1$$

Math-2 37h, D1h (Add)

37h =	0011 0111	AF = 0 GF = 1 PF = 0 SF = 0 ZF = 0 OF = 0
D1h = (+)	1101 0001	
1 0000 1000		
← carry फिर आया है		

Math-3 9Eh, D3h (Add)

9Eh =	1001 1110	AF = 1 CF = 1 PF = 1 SF = 0 ZF = 0 OF = 1
D3h = (+)	1101 0011	
1 0111 0001		
!		

Math-4: Analyze flag bits (CF, SF, ZF, PF, OF) after executing the following instructions.

MOV BL, CL
 ADD AL, BL
 NEG AL
 ↓
 2's complement

Given,

AL = B5h
 CL = CAL

instruction
 on slide page
 11

Solⁿ: For MOV, CL no flags are affected.

B5h, CAL (Add)

$$\begin{array}{r} \text{B5h} = \quad 1011 \ 0101 \\ \text{CAL} = (+) \quad 1100 \ 1010 \\ \hline \quad 10111 \ 1111 \\ \vdots \end{array} \quad \left| \begin{array}{l} \text{CF} = 1 \\ \text{SF} = 0 \\ \text{ZF} = 0 \\ \text{PF} = 0 \\ \text{OF} = 1 \end{array} \right.$$

$$\begin{array}{r} \text{AL} = 0111 \ 1111 \\ \text{NEG AL} = 10000001 \end{array} \quad \left| \begin{array}{l} \text{CF} = 1 \\ \text{SF} = 1 \\ \text{ZF} = 0 \\ \text{PF} = 1 \\ \text{OF} = 0 \end{array} \right.$$

Note: যদি instructions serially থাকে কোনো number করা না থাকে তাহলে একটি অন্যটির সাথে connected. ক

আর যদি number দেওয়া থাকে তাহলে individually solve করতে হবে individually value consider করে।